

FIN-02: Cost Structure — Project AEGIS / Carrington Storm Motors

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Executive Summary: The Cost Thesis

Carrington Storm Motors’ cost structure is designed around three principles: (1) asset-light manufacturing in Years 1-3, transitioning to vertically integrated production in Years 4-5, (2) R&D intensity that front-loads innovation before competitors enter, and (3) a headcount model that scales revenue 25x (from \$11.5M to \$1.2B) while scaling employees 36x (from 25 to 900) — improving revenue-per-employee from \$639K to \$1.78M.

Williams Heuristic: We stockpile knowledge cheaply (6,302 documents cost far less than building factories), then convert that knowledge into premium-priced products. The margin structure reflects this: low initial margins as we build manufacturing capability, expanding margins as volume and the learning curve compound.

Cost of Goods Sold (COGS) — By Product Line

TABLE 2.1: COGS Breakdown by Product Line (\$M USD)

Aegis-C Composite Shielding (ACS)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of ACS Revenue (Y5)
Raw Materials (polymers, dielectric fillers, conductors)	\$0.8	\$3.0	\$11.2	\$30.0	\$60.0	20.0%
Direct Labor (composite techs, QC)	\$0.5	\$1.8	\$5.6	\$15.0	\$30.0	10.0%
Manufacturing Overhead (amortized line cost, energy, facilities)	\$0.3	\$1.1	\$3.5	\$9.6	\$24.0	8.0%
Packaging, Shipping, Installation Support	\$0.1	\$0.4	\$1.4	\$4.8	\$12.0	4.0%
Warranty Reserve (2% of revenue)	\$0.04	\$0.16	\$0.7	\$2.4	\$6.0	2.0%
Total ACS COGS	\$1.74	\$6.46	\$22.4	\$61.8	\$132.0	44.0%
ACS Gross Profit	\$0.26	\$1.54	\$12.6	\$58.2	\$168.0	—

COGS Component	Y1	Y2	Y3	Y4	Y5	% of ACS Revenue (Y5)
ACS Gross Margin %	13.0%	19.3%	36.0%	48.5%	56.0%	—

Accountant Heuristic: Y1 margin is thin because initial production is manual (high labor per unit) and material purchases are in R&D quantities (no volume discounts). By Y5, the automated layup lines produce at 5 units/day with material costs 40% lower due to bulk procurement, and labor per unit drops 85% from automation. This is a classic manufacturing learning curve — every doubling of cumulative volume reduces unit cost by 15-18%.

Aegis Home (AHM)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of AHM Revenue (Y5)
Raw Materials (composite panel, electronics, enclosure)	\$0.15	\$0.90	\$6.0	\$20.0	\$50.4	18.0%
Direct Labor (assembly, testing, packaging)	\$0.08	\$0.45	\$3.0	\$10.0	\$25.2	9.0%
Manufacturing Overhead	\$0.06	\$0.30	\$1.75	\$6.0	\$16.8	6.0%
Packaging, Shipping, Installation	\$0.04	\$0.21	\$1.5	\$5.0	\$14.0	5.0%
Support						
Warranty Reserve (2%)	\$0.01	\$0.06	\$0.5	\$2.0	\$5.6	2.0%
Total AHM COGS	\$0.34	\$1.92	\$12.75	\$43.0	\$112.0	40.0%
AHM Gross Profit	\$0.16	\$1.08	\$12.25	\$57.0	\$168.0	—
AHM Gross Margin %	32.0%	36.0%	49.0%	57.0%	60.0%	—

El Segundo Calm: The home kit is a box of pre-cut composite panels, a set of clamping hardware, and an installation guide. It costs less to make than a mid-range dishwasher and sells for more. The margin structure reflects this: it's a manufactured product with strong IP, not a commodity. By Y5, the kit costs ~\$1,436 to produce and sells for \$3,590 — a 60% gross margin that compares favorably to consumer electronics (Apple: 43%, Garmin: 58%, Sonos: 46%).

Charlemagne-Class Fleet (CCF)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of CCF Revenue (Y5)
Hull & Propulsion (subcontracted naval construction)	\$1.8	\$6.1	\$21.1	\$48.0	\$108.5	31.0%
EMP Restoration Equipment (transformers, switchgear, cable)	\$1.5	\$5.0	\$16.9	\$37.5	\$84.0	24.0%
C4ISR Systems (satcom, hardened computing)	\$0.5	\$1.8	\$6.5	\$15.0	\$38.5	11.0%
Integration Labor & Engineering	\$0.5	\$1.7	\$5.2	\$12.0	\$28.0	8.0%
Sea Trials, Certification, Delivery	\$0.3	\$0.9	\$3.3	\$7.5	\$17.5	5.0%
Warranty & Post-Delivery Support (3%)	\$0.15	\$0.54	\$1.95	\$4.5	\$10.5	3.0%
Total CCF COGS	\$4.75	\$16.04	\$54.95	\$124.5	\$287.0	82.0%
CCF Gross Profit	\$0.25	\$1.96	\$10.05	\$25.5	\$63.0	—
CCF Gross Margin %	5.0%	10.9%	15.5%	17.0%	18.0%	—

Williams Heuristic: Ships are expensive. Big metal boxes that float, filled with even more expensive electrical gear. We're not building iPhones. The margin on vessel contracts is defense-industry standard (Huntington Ingalls: 14-16%, General Dynamics Marine: 8-12%). Our margin improves as the supply chain for EMP-hardened equipment matures and we move from custom one-offs to class-production. The first Charlemagne takes 18 months and costs almost everything we charge; the 48th takes 10 months and earns 18%. That's shipbuilding.

Phantom MK-1 Robotics (PMR)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of PMR Revenue (Y5)
Hardware Components (motors, sensors, chassis, composites)	\$0.35	\$1.55	\$5.4	\$13.2	\$26.4	22.0%
Assembly & Testing Labor	\$0.15	\$0.65	\$2.0	\$4.4	\$7.2	6.0%
Manufacturing Overhead	\$0.08	\$0.35	\$1.2	\$2.75	\$6.0	5.0%

COGS Component	Y1	Y2	Y3	Y4	Y5	% of PMR Revenue (Y5)
Software Development (capitalized per unit)	\$0.05	\$0.20	\$0.8	\$2.2	\$4.8	4.0%
Packaging, Shipping, Training	\$0.03	\$0.15	\$0.6	\$1.65	\$3.6	3.0%
Warranty Reserve (2%)	\$0.02	\$0.10	\$0.4	\$1.1	\$2.4	2.0%
Total PMR COGS	\$0.68	\$3.00	\$10.4	\$25.3	\$50.4	42.0%
PMR Gross Profit	\$0.32	\$2.00	\$9.6	\$29.7	\$69.6	—
PMR Gross Margin %	32.0%	40.0%	48.0%	54.0%	58.0%	—

Infrastructure Hardening (IFH)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of IFH Revenue (Y5)
Materials (composites, grounding systems, shielding)	\$0.7	\$2.6	\$7.5	\$15.4	\$28.6	26.0%
Direct Installation Labor	\$0.5	\$1.9	\$5.3	\$10.5	\$19.8	18.0%
Engineering & Project Management	\$0.2	\$0.7	\$2.0	\$4.4	\$8.8	8.0%
Equipment & Tooling	\$0.1	\$0.4	\$1.0	\$2.2	\$3.3	3.0%
Permits, Inspections, Compliance	\$0.06	\$0.24	\$0.75	\$1.65	\$3.3	3.0%
Warranty Reserve (2%)	\$0.04	\$0.16	\$0.5	\$1.1	\$2.2	2.0%
Total IFH COGS	\$1.60	\$6.00	\$17.05	\$35.25	\$66.0	60.0%
IFH Gross Profit	\$0.40	\$2.00	\$7.95	\$19.75	\$44.0	—
IFH Gross Margin %	20.0%	25.0%	31.8%	35.9%	40.0%	—

Insurance Premium Partnerships (IPP)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of IPP Revenue (Y5)
Actuarial Staff & Data Infrastructure	\$0.1	\$0.3	\$1.0	\$1.5	\$2.4	8.0%
Partner Integration & Onboarding	\$0.05	\$0.2	\$0.6	\$0.9	\$1.5	5.0%
Claims Verification Operations	\$0.01	\$0.04	\$0.16	\$0.3	\$0.6	2.0%

COGS Component	Y1	Y2	Y3	Y4	Y5	% of IPP Revenue (Y5)
Total IPP COGS	\$0.16	\$0.54	\$1.76	\$2.7	\$4.5	15.0%
IPP Gross Profit	\$0.34	\$1.46	\$6.24	\$12.3	\$25.5	—
IPP Gross Margin %	68.0%	73.0%	78.0%	82.0%	85.0%	—

Licensing & IP Royalties (LIR)

COGS Component	Y1	Y2	Y3	Y4	Y5	% of LIR Revenue (Y5)
IP Legal & Maintenance (patent filings, enforcement)	\$0.08	\$0.13	\$0.20	\$0.35	\$0.50	5.0%
Technology Transfer & Support	\$0.04	\$0.07	\$0.10	\$0.15	\$0.20	2.0%
Total LIR COGS	\$0.12	\$0.20	\$0.30	\$0.50	\$0.70	7.0%
LIR Gross Profit	\$0.38	\$0.80	\$1.70	\$4.50	\$9.30	—
LIR Gross Margin %	76.0%	80.0%	85.0%	90.0%	93.0%	—

TABLE 2.2: Consolidated COGS and Gross Margin (\$M USD)

Metric	Y1	Y2	Y3	Y4	Y5
Total Revenue	\$11.50	\$45.00	\$180.00	\$500.00	\$1,200.00
Total COGS	\$9.39	\$34.16	\$119.60	\$293.05	\$652.60
Total Gross Profit	\$2.11	\$10.84	\$60.40	\$206.95	\$547.40
Blended Gross Margin %	18.3%	24.1%	33.6%	41.4%	45.6%

El Segundo Calm: Blended margin starts low because the Y1 revenue mix is heavy on low-margin vessel contracts and light on high-margin insurance/IP. As the mix shifts (insurance from 4% to 2.5% of revenue, but IPP+LIR from 9% to 3.3% of revenue — counterintuitively, they shrink as a share but grow in absolute profit dollars), and as the manufacturing learning curve kicks in on the hardware lines, blended margin approaches 46%. For an industrial technology company, 45%+ blended gross margin is excellent (compare: Tesla 18%, General Electric 26%, Raytheon 29%, ASML 51%).

Operating Expenses

Research & Development (R&D)

What It Covers: Materials science research (ongoing refinement of composite dielectric formulations), product engineering (new product variants, next-gen Phantom MK-2, composite application automation), testing and certification (Z-Machine shots, White Sands EMP sim testing, UL/ETL certification), and the maintenance/expansion of the research corpus.

R&D Component	Y1	Y2	Y3	Y4	Y5
Materials Science Team (headcount)	6	12	28	55	95
Product Engineering Team (headcount)	4	10	24	50	90
Testing & Certification	\$1.2M	\$2.5M	\$5.0M	\$8.0M	\$12.0M
Lab Facilities & Equipment	\$0.6M	\$1.5M	\$3.5M	\$6.0M	\$10.0M
Research Corpus Maintenance	\$0.2M	\$0.3M	\$0.5M	\$0.8M	\$1.0M
Total R&D Expense	\$3.5M	\$8.0M	\$22.0M	\$48.0M	\$90.0M
R&D as % of Revenue	30.4%	17.8%	12.2%	9.6%	7.5%

Accountant Heuristic: R&D intensity starts high (30% of revenue) because we're a pre-revenue research organization transitioning to commercial. It declines as a percentage while growing 25x in absolute dollars. This is the standard pattern for deep-tech companies. The 6,302-document corpus is the crystallized R&D spend that predates this model — representing an estimated \$15-25M of pre-Series A investment (founder sweat equity, university research partnerships, grant funding).

Sales & Marketing (S&M)

S&M Component	Y1	Y2	Y3	Y4	Y5
Government BD Team (headcount)	3	8	18	30	45
Commercial Sales Team (headcount)	2	6	16	35	65
Marketing & Communications	\$0.3M	\$1.0M	\$3.0M	\$8.0M	\$18.0M
Trade Shows & Conferences	\$0.2M	\$0.5M	\$1.0M	\$2.0M	\$3.0M
Insurance Channel Development	\$0.2M	\$0.8M	\$2.5M	\$5.0M	\$8.0M
International BD Travel (200-country network)	\$0.3M	\$0.8M	\$2.0M	\$4.0M	\$6.0M
Certified Installer Training Program	\$0.1M	\$0.4M	\$1.5M	\$3.0M	\$5.0M
Total S&M Expense	\$2.0M	\$6.5M	\$22.0M	\$58.0M	\$120.0M
S&M as % of Revenue	17.4%	14.4%	12.2%	11.6%	10.0%

General & Administrative (G&A)

G&A Component	Y1	Y2	Y3	Y4	Y5
Executive Leadership (18-Agent Command)	\$1.5M	\$3.0M	\$5.0M	\$8.0M	\$12.0M
Finance, Legal, HR (headcount)	3	10	25	55	100
Headquarters & Facilities	\$0.5M	\$1.5M	\$3.5M	\$8.0M	\$18.0M
IT & Security Infrastructure	\$0.3M	\$0.8M	\$2.0M	\$5.0M	\$10.0M
Insurance (corporate, D&O, cyber)	\$0.2M	\$0.5M	\$1.5M	\$3.5M	\$6.0M
Professional Services (audit, legal, consulting)	\$0.3M	\$0.8M	\$2.0M	\$5.0M	\$10.0M
Travel, Miscellaneous	\$0.2M	\$0.4M	\$1.0M	\$2.5M	\$4.0M

G&A Component	Y1	Y2	Y3	Y4	Y5
Total G&A Expense	\$3.0M	\$7.0M	\$15.0M	\$32.0M	\$60.0M
G&A as % of Revenue	26.1%	15.6%	8.3%	6.4%	5.0%

Note on the 18-Agent Command: CSM’s governance structure includes 18 senior leaders (CEO, CTO, CFO, CRO, COO, Chief Science Officer, Chief Policy Officer, 7 division VPs, 3 chief engineers, and General Counsel). This is deliberately lean for a company targeting \$1.2B revenue. Average executive comp is \$670K Y5 (salary + bonus + equity), consistent with pre-IPO industrial tech companies.

TABLE 2.3: Consolidated Operating Expenses (\$M USD)

Expense Category	Y1	Y2	Y3	Y4	Y5
R&D	\$3.5	\$8.0	\$22.0	\$48.0	\$90.0
S&M	\$2.0	\$6.5	\$22.0	\$58.0	\$120.0
G&A	\$3.0	\$7.0	\$15.0	\$32.0	\$60.0
Total OpEx	\$8.5	\$21.5	\$59.0	\$138.0	\$270.0
OpEx as % of Revenue	73.9%	47.8%	32.8%	27.6%	22.5%
EBITDA	(\$6.39)	(\$10.66)	\$1.40	\$68.95	\$277.40
EBITDA Margin %	-55.6%	-23.7%	0.8%	13.8%	23.1%

Headcount Growth Projections

TABLE 2.4: Headcount by Function (Period-End)

Function	Y1	Y2	Y3	Y4	Y5
R&D (Materials Science + Engineering)	10	22	52	105	185
Manufacturing & Operations	6	18	55	140	320
Sales & Business Development	5	14	34	65	110
Marketing & Communications	1	4	12	25	45
G&A (Finance, Legal, HR, IT, Exec)	8	20	47	90	160
Field Operations (Installation, Training)	0	5	18	45	80
Total Headcount	25	75	200	450	900

TABLE 2.5: Compensation Structure

Level	Y1 Avg. Comp	Y3 Avg. Comp	Y5 Avg. Comp	% with Equity
C-Suite (Exec)	\$350K	\$500K	\$670K	100%
VP / Director	\$220K	\$300K	\$400K	100%
Senior Engineer / Manager	\$150K	\$185K	\$220K	80%
Engineer / Specialist	\$110K	\$135K	\$160K	60%
Technician / Support	\$65K	\$80K	\$95K	30%
Blended Average (all employees)	\$132K	\$158K	\$178K	52%

TABLE 2.6: Total Personnel Cost (\$M USD)

Component	Y1	Y2	Y3	Y4	Y5
Salary + Bonus	\$2.3	\$7.5	\$21.0	\$51.0	\$108.0
Benefits + Payroll Tax (25% load)	\$0.6	\$1.9	\$5.3	\$12.8	\$27.0
Equity Compensation (non-cash)	\$0.4	\$1.5	\$5.0	\$14.0	\$35.0
Total Personnel Cost	\$3.3	\$10.9	\$31.3	\$77.8	\$170.0
Personnel as % of OpEx	38.8%	50.7%	53.1%	56.4%	63.0%
Personnel as % of Revenue	28.7%	24.2%	17.4%	15.6%	14.2%

Manufacturing Capacity & Capital Expenditure

TABLE 2.7: Production Capacity Buildout

Facility	Y1	Y2	Y3	Y4	Y5
Composite Layup Lines	0 (manual)	1 semi-auto	3 automated	5 automated	8 automated
ACS Annual Capacity (units)	120	500	2,000	6,000	14,000
AHM Assembly Lines	0 (3rd party)	1	2	4	7
AHM Annual Capacity (kits)	300	1,500	10,000	35,000	85,000
Phantom Assembly	Manual bench	1 line	2 lines	3 lines	5 lines
PMR Annual Capacity (systems)	30	120	450	1,100	2,200
Total Manufacturing Sq. Ft.	5,000	15,000	45,000	90,000	160,000

TABLE 2.8: Capital Expenditure (\$M USD)

CapEx Category	Y1	Y2	Y3	Y4	Y5
Manufacturing Equipment (automated lines)	\$0.5	\$3.0	\$8.0	\$15.0	\$25.0
Lab Equipment & Testing Facilities	\$1.0	\$2.5	\$5.0	\$8.0	\$12.0
Facility Buildout (leasehold improvements)	\$0.3	\$1.5	\$3.0	\$5.0	\$8.0
IT Infrastructure & Software	\$0.2	\$0.5	\$1.5	\$3.0	\$5.0
Fleet Prototype Tooling	\$1.0	\$2.0	\$3.0	\$4.0	\$6.0
Total CapEx	\$3.0	\$9.5	\$20.5	\$35.0	\$56.0

Depreciation & Amortization Schedule

Category	Useful Life	Annual Depreciation Method
Manufacturing Equipment	7 years	Straight-line

Category	Useful Life	Annual Depreciation Method
Lab Equipment	5 years	Straight-line
Facility Leasehold Improvements	10 years	Straight-line
IT Hardware	3 years	Straight-line
Software (capitalized development)	3 years	Straight-line
IP / Intangible Assets	10 years	Straight-line

Projected D&A (\$M): Y1: \$0.5, Y2: \$1.5, Y3: \$3.5, Y4: \$7.0, Y5: \$12.0

Key Cost Ratios and Benchmarks

Ratio	Y1	Y2	Y3	Y4	Y5	Industry Benchmark
Gross Margin	18.3%	24.1%	33.6%	41.4%	45.6%	25-50% (Industrial Tech)
R&D / Revenue	30.4%	17.8%	12.2%	9.6%	7.5%	5-15%
S&M / Revenue	17.4%	14.4%	12.2%	11.6%	10.0%	10-20%
G&A / Revenue	26.1%	15.6%	8.3%	6.4%	5.0%	5-12%
OpEx / Revenue	73.9%	47.8%	32.8%	27.6%	22.5%	20-35%
Revenue / Employee (\$K)	\$639	\$900	\$1,304	\$1,538	\$1,778	\$300-800K
Personnel / OpEx	38.8%	50.7%	53.1%	56.4%	63.0%	50-70%

Cost Optimization Levers (Already Modeled)

1. **Learning Curve:** 15% cost reduction per doubling of cumulative production volume for ACS and AHM; 10% for PMR and CCF (more engineering-heavy).
2. **Vertical Integration:** Y1-Y2 rely on contract manufacturing for AHM enclosures and Phantom chassis. By Y3-Y4, insourcing these components improves margin 3-5 points.
3. **Bulk Material Procurement:** By Y5, raw material purchasing power reduces input costs 25-40% vs. Y1 R&D-quantity pricing.
4. **Automation:** Each automated composite layup line replaces 8-12 manual technicians. Capital cost amortizes over 7 years at ~\$500K/year/line, vs. \$800K-1.2M/year in labor saved. Net savings per line: \$300-700K/year.
5. **Geographic Labor Arbitrage:** Y4+ manufacturing expansion considers lower-cost US regions (Appalachia, Rust Belt repurposed factories) and allied-nation production (UK, Australia) with defense offset benefits.

End of FIN-02: Cost Structure — Carrington Storm Motors, Project AEGIS Next: FIN-03: P&L Projection